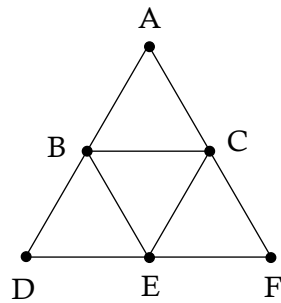


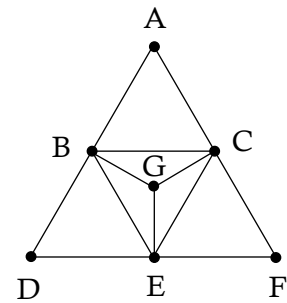
Ore's Theorem

1. Consider the graphs given below.
 - i. Determine directly whether the graph satisfies the condition of Ore's Theorem, justifying your answer.
 - ii. For the graphs which do satisfy the condition of Ore's Theorem, write down a Hamiltonian cycle.
 - iii. For the graphs which do not satisfy the condition of Ore's Theorem, determine whether they contain a Hamiltonian cycle.

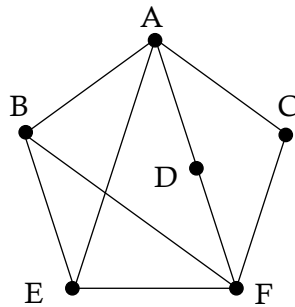
a.



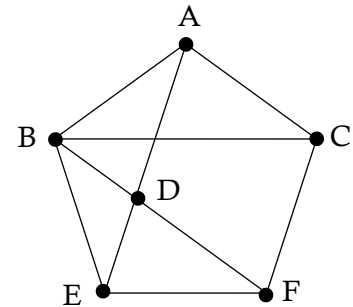
b.



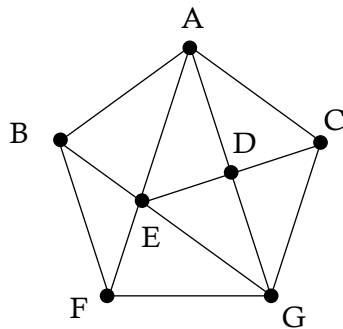
c.



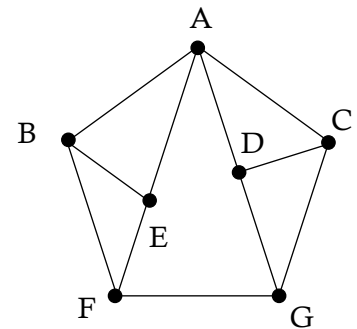
d.



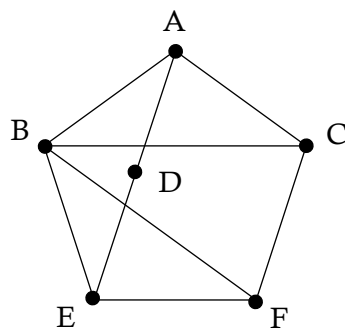
e.



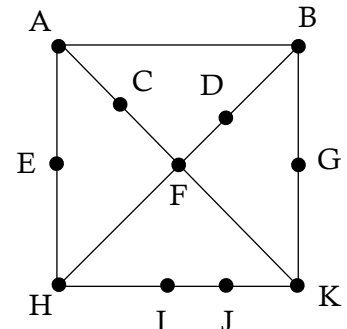
f.



g.



h.



2.
 - a. Explain why no graph with degree sequence 4, 4, 4, 3, 3, 3, 3 can satisfy the conditions of Ore's Theorem.
 - b. Draw a graph with degree sequence 4, 4, 4, 3, 3, 3, 3 that contains a Hamiltonian cycle.
 - c. Draw a simple graph with degree sequence 4, 4, 4, 3, 3, 3, 3 that does not contain a Hamiltonian cycle. Justify how you know that your graph does not contain a Hamiltonian cycle.

3.
 - a. A simply-connected graph has order 5 and size 8. Use Ore's Theorem to prove that the graph contains a Hamiltonian cycle.
 - b. Provide an example of a simply-connected graph of order 5 and size 7 such that:
 - i. the graph satisfies the condition of Ore's Theorem;
 - ii. the graph is Hamiltonian, but does not satisfy the condition of Ore's Theorem;
 - iii. the graph is not Hamiltonian.

4. Suppose that a simple graph G satisfies the condition that the sum of the degrees of any two non-adjacent vertices is at least $|G| - 1$ (that is, it just misses out on satisfying the condition of Ore's Theorem).
Prove that G must be connected.

5. Find a simple graph of order 6 which does not contain a Hamiltonian cycle, and in which the sum of the degrees of *every* pair of non-adjacent vertices is equal to 5.

6. For a simple graph G , generalise the result of Question 3 to determine the maximum number of edges in a non-Hamiltonian graph of order n . Equivalently, find the minimum number of edges required to guarantee that a graph of order n contains a Hamiltonian cycle.